SPI Design & UVM Verification - Master Project Index

1. Project Architecture Overview

This directory contains detailed technical explanations and interview-ready documentation for every single file in the SPI Project.

The project implements a high-performance, fully parameterized Serial Peripheral Interface (SPI) Master and Slave architecture in Verilog/SystemVerilog, accompanied by a comprehensive UVM 1.2 Verification Environment in AMD Vivado 2025.1.

Key Features: • Parameterized DATA_WIDTH (8, 16, 32-bit transfers) • Configurable Clock Frequency via programmable system clock divider (CLK_DIV) • Full hardware support for all 4 SPI Modes (CPOL=0/1, CPHA=0/1) • High-Z Tri-Stated MISO bus when CS_N is deasserted (multi-peripheral compliant) • Simultaneous full-duplex bi-directional transmission verification • Layered UVM environment with constrained-random stimulus and functional coverage.

2. RTL Design Files

1. 01_spi_top.pdf: Top-level hardware wrapper interconnecting Master and Slave. 2. 02_spi_master.pdf: Parameterized Master controller with precise CS_N framing and FSM. 3. 03_spi_slave.pdf: Reactive Slave peripheral with tri-stated MISO and synchronous sampling.

3. UVM Verification Components

4. 04_spi_interface.pdf: SystemVerilog interface bundle with clocking blocks. 5. 05_spi_transaction.pdf: Sequence item containing randomized payloads and mode controls. 6. 06_spi_sequence.pdf: Directed 4-mode sequences, corner vectors, and random generator. 7. 07_spi_sequencer.pdf: UVM sequence-to-driver communication arbiter. 8. 08_spi_driver.pdf: Pin-level protocol driver handling handshakes and mode settling. 9. 09_spi_monitor.pdf: Dual-channel passive bus sniffer. 10. 10_spi_coverage.pdf: Functional coverage subscriber covering CPOLxCPHA and data ranges. 11. 11_spi_scoreboard.pdf: Self-checking dual FIFO comparator validating MOSI & MISO. 12. 12_spi_agent.pdf: Reusable verification agent encapsulating sequencer, driver, monitor, and coverage. 13. 13_spi_env.pdf: Top verification environment wiring agent analysis ports to scoreboard. 14. 14_spi_test.pdf: Test library (spi_test, spi_rand_test, spi_modes_test). 15. 15_spi_tb.pdf: Simulation top module with 100MHz clock, reset, and waveform dumping.

4. Comprehensive Reports

• SPI_Project_Report.pdf: Complete technical project overview and synthesis guide. • SPI_Master_Interview_Report.pdf: Deep-dive interview questions and answers on SPI protocols and verification. • SPI_Waveform_Analysis_Guide.pdf: Timing diagrams and waveform analysis across all 4 modes.